Paragraph 1

Retrieval-Augmented Generation (RAG) is an artificial intelligence technique that combines information retrieval with large language models. Instead of relying only on the model's internal knowledge, RAG systems retrieve relevant information from external documents before generating an answer. This approach significantly improves accuracy and reduces hallucinations.

Paragraph 2

In a typical RAG pipeline, documents are first loaded and processed. The text is then split into smaller chunks to preserve context and improve retrieval quality. Each chunk represents a meaningful portion of text that can be independently searched and analyzed during the retrieval process.

Paragraph 3

Once the text is chunked, embeddings are generated for each chunk. Embeddings are numerical representations that capture the semantic meaning of the text. These embeddings allow the system to perform semantic search, meaning it can find relevant content even if the exact words do not match the user's query.

Paragraph 4

When a user asks a question, the system converts the query into an embedding and compares it with the stored document embeddings. Using similarity measures such as cosine similarity, the system retrieves the most relevant chunks. These retrieved chunks form the context that is passed to the language model for answer generation.

Paragraph 5

The language model then generates a response using only the retrieved context. This ensures that the answer is grounded in the original documents. If the relevant information is not found in the documents, a well-designed RAG system should clearly indicate that the answer is not available.

Paragraph 6

RAG systems are widely used in applications such as document question answering, chatbots, knowledge bases, and enterprise search systems. They are particularly useful when working with private or constantly changing data, where relying solely on pre-trained model knowledge is insufficient.